Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is a branch of computer science that focuses on building systems capable of performing tasks that normally require human intelligence. These tasks include reasoning, problem-solving, learning, decision-making, language understanding, and image recognition.

Machine Learning (ML) is a subset of AI that enables computers to learn from data without being explicitly programmed. Instead of following fixed rules, machine learning algorithms identify patterns in data and use those patterns to make predictions or decisions.

There are three major types of machine learning:

1. **Supervised Learning**
    - Uses labeled data.
    - Examples include spam email detection and house price prediction.
2. **Unsupervised Learning**
    - Uses unlabeled data.
    - Examples include customer segmentation and clustering.
3. **Reinforcement Learning**
    - Learns by interacting with an environment using rewards and penalties.
    - Examples include robotics and game-playing AI.

Natural Language Processing (NLP) is a field of AI that enables computers to understand and generate human language. Applications include chatbots, virtual assistants, translation systems, and text summarization.

Large Language Models (LLMs), such as Llama and GPT, are trained on massive amounts of text data. They can answer questions, summarize documents, generate code, and assist users in many different tasks.

Embeddings are numerical vector representations of text. They capture the semantic meaning of words, sentences, or documents. Similar pieces of text produce similar embeddings, making them useful for semantic search and retrieval.

Vector databases such as ChromaDB store embeddings efficiently. Instead of searching using exact keywords, vector databases compare embeddings using similarity metrics such as cosine similarity to find the most relevant information.

Retrieval-Augmented Generation (RAG) combines document retrieval with language models. First, relevant document chunks are retrieved from a vector database. Then,

those chunks are provided as context to an LLM, allowing it to generate accurate and context-aware responses.

Ollama enables developers to run open-source language models locally on their own computers. This improves privacy, reduces dependency on cloud services, and allows offline AI applications.

ChromaDB is an open-source vector database designed for AI applications. It supports storing document embeddings, performing similarity searches, filtering by metadata, and integrating with local language models such as those provided by Ollama.

AI, machine learning, vector databases, embeddings, and RAG together form the foundation of modern intelligent applications such as document question answering, semantic search, AI assistants, and enterprise knowledge retrieval systems.